Jamie Le

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Education

University of Maryland College Park, Maryland

Math Major, Minor in Quantum Science and Engineering Sept 2023 - May 2027 *(expected)*

- Current GPA: 3.8/4.0

Research

Murmurations of ${}_3F_2$ Hypergeometric Families

Polymath Jr. Undergraduate Research Project June 2026 - Current

- Studied recent literature on murmurations arising from averages of hypergeometric function values over elliptic curves.
- Explored connections between representation theory and arithmetic geometry.
- Contributing to a research manuscript and investigating a more explicit expression for the local factor in the GL(3) case.

Algebraic Cryptanalysis of the SNOVA Signature Scheme

MathQuantum Undergraduate Research Fellowship Jan 2026 - April 2026

- Investigated algebraic structures underlying post-quantum signature schemes, with a focus on understanding the mathematical foundations of the SNOVA construction.
- Worked under Prof. Carl Miller to analyze structural properties and algorithmic behavior.

Extensions of Quantum Developments to Classical Automata

Undergraduate Quantum Computing Research Project Jan 2026 - May 2026

- Investigated quantum analogues of classical computational models, including quantum finite automata, probabilistic finite automata, and quantum pushdown automata, through a review of the literature.

Mathematical Interests

- Number Theory
- Arithmetic Geometry
- Quantum Computing

Relevant Coursework

MATH403: Introduction to Abstract Algebra A (Fall 2025)

- Proof-based study of groups, homomorphisms, quotient structures, and introductory ring theory

MATH404: Field Theory B (Spring 2026)

- Proof-based study of rings, ideals, fields, field extensions, algebraic closures, and Galois theory

MATH406: Introduction to Number Theory A- (Fall 2024)

- Modular arithmetic, classical number-theoretic theorems, and quadratic reciprocity

MATH405: Linear Algebra B (Spring 2026)

- A proof-based study of topics such as vector spaces, linear operators, the dot product, and eigenvalue decomposition

STAT410: Introduction to Probability Theory A+ (Summer 2025)

- Probability spaces, random variables, expectation, and foundational limit theorems

Additional Coursework: MATH462 (Partial Differential Equations, A+), MATH410 (Advanced Calculus I, B+), PHYS360 (Introduction to Quantum Mechanics, A), and PHYS457 (Introduction to Quantum Computing, B+)

Courses in Progress: MATH600 (Abstract Algebra I, Audit), MATH407 (Advanced Number Theory), MATH411 (Advanced Calculus II)

Presentations & Expository Work

Representing Quantum Operators and States
MathQuantum High School Summer Program July 2026

- Delivered a guest lecture to high school MathQuantum summer fellows on the Spectral Theorem and Schmidt Decomposition, emphasizing mathematical understanding through proofs and exercises.
- Prepared a LaTeX slide deck using Overleaf to support an hour-long lecture.

A First Look at the Theory Behind Calculus
Undergraduate Expository Mathematics Project Dec 2025 (revised in Sept 2026)

- Interviewed Math Department Advisor Dr. Ida Chan and a Math + Computer Science double major to identify challenges students face in transitioning to proof-based mathematics.
- Designed a theory-focused Calculus I–II handout to help students transition toward proof-based mathematics

Exploring Infinity: The Beauty and Structure of Ordinals
Presentation as part of the Direct Reading Program at UMD Sept 2024 - Dec 2024

- Collaborated weekly with a graduate mentor on a reading project in Set Theory and Ordinals
- Designed and delivered a Google Slides presentation on findings to an audience of graduate and undergraduate students

Teaching

Grader for Introduction to Number Theory (MATH406) College Park, Maryland
Department of Mathematics Sept 2025 - Dec 2025

- Graded proof-based homework and provided detailed written feedback
- Communicated recurring conceptual difficulties to the course instructor

Tutor for Linear Algebra (MATH240/461) College Park, Maryland
Department of Mathematics Sept 2026 - Current

- Provide tutoring and conceptual support to undergraduate students in linear algebra.

Calculus 2 and Calculus 3 Tutor College Park, Maryland
Varsity Tutors Sept 2025 - Dec 2025

- Tutored undergraduate mathematics including Calculus 2 and 3

Awards and Honors

- **MathQuantum Undergraduate (NSF Stipend) Fellowship**, University of Maryland Spring 2026
- **Dean's Honor**, University of Maryland Fall 2023, Spring 2024, Fall 2024, and Fall 2025
- **Maryland Vietnamese Mutual Association Academic Achievement and Community Service Award**, Montgomery County Public Schools Spring 2023

Conferences & Workshops Attended

2026 Joint Mathematics Meeting Washington, D.C.
Conference at the Walter E. Washington Convention Center Jan 2026

- Attended research talks and special sessions in Abstract Algebra and Number Theory

Mathematics-Opportunities in Research and Education Blacksburg, Virginia
Conference at Virginia Tech Apr 2025

- Participated in sessions on graduate programs and undergraduate research opportunities

Quantum.Tech USA 2025 Washington, D.C.
Conference at the Conrad Hotel Apr 2025

- Represented the Q^3 Initiative and engaged with academic and industry participants

Skills

Languages: English and Vietnamese.

Programming: `Python`, `C`, `HTML`, and `MATLAB`.

Mathematical Tools: Overleaf and LaTeX

Additional Activities

Logistics Lead Organizer College Park, Maryland

Q^3 Initiative (Quantum and Art Hackathon) Jan 2025 - Sept 2025

- Led logistics coordination for a multidisciplinary quantum and art hackathon, managing communication and task execution

Student Volunteer College Park, Maryland

MathQuantum (@ UMD Maryland Day Event) April 2026

- Guided Maryland Day participants through quantum themed puzzles.

Discussion Leader College Park, Maryland

Math Club Reading Group of 4 Students Jan 2025 - April 2025

- Co-led a small reading group reading papers in model theory, algebra, and number theory

Test Grader and Proctor College Park, Maryland

MMATHS Competition Oct 2024 & Nov 2025

- Proctored exams and graded submissions for a regional mathematics competition

Additional Coursework

LPS Quantum Computing Summer Short Course Completed (June 2026)

- Completed a 2-week summer short course in the fundamentals of quantum computing concerning quantum algorithms and materials science